

The Training will begin shortly

Architecting Agentic AI Business Solutions with Microsoft AI

Gene Mckelvey
Netcom Learning

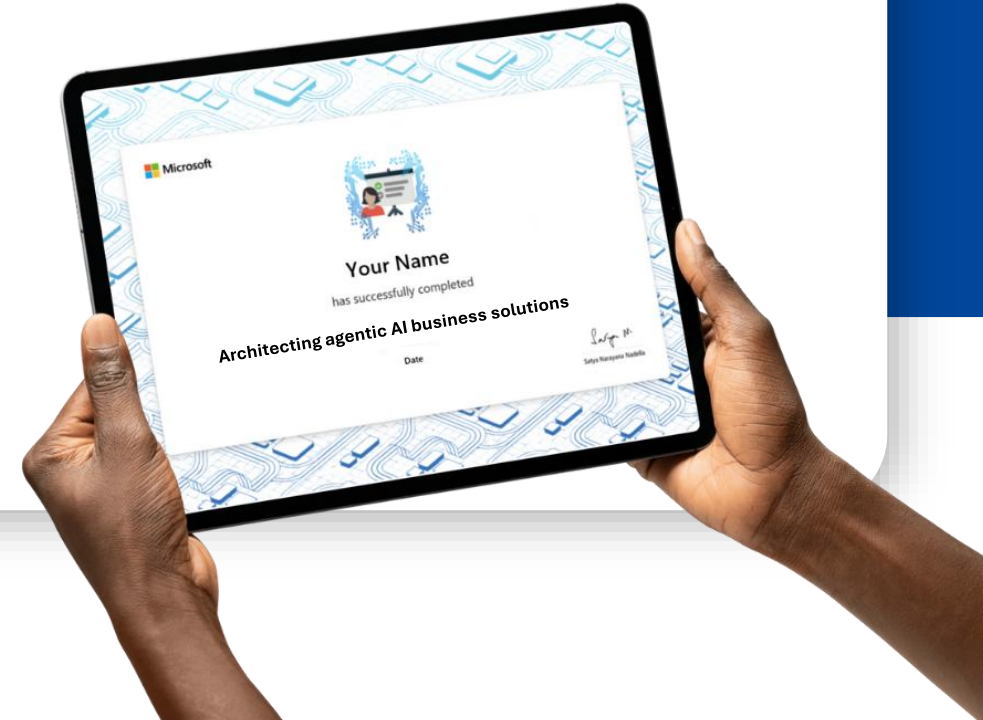
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**Hello,
Thank you for joining me Today!**

Gene Mckelvey
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Welcome & Today's Journey

What Agentic AI really is

How Microsoft enables enterprise AI

Designing business-ready solutions

Security & governance

Enterprise architecture

Real-world demonstrations

TODAY'S JOURNEY

Understand



Design



Build



Govern



Scale

AGENDA

01

Understanding Agentic AI

From automation to autonomous, goal-driven business action

02

The Microsoft AI Ecosystem

Copilot, Copilot Studio, Power Platform, Dynamics 365, Azure AI & Fabric

03

Architecting Enterprise Solutions

Reference architecture, design patterns and grounding with enterprise data

04

Governance, Security & Scaling

Responsible AI, security layers, lifecycle governance and the 90-day roadmap

SECTION 01

Understanding Agentic AI



Why Agentic AI Matters Now



Business Impact

- Less manual work
- Faster decision-making
- Higher productivity
- Continuous, always-on operations
- Better customer experiences

BUSINESS EXAMPLE

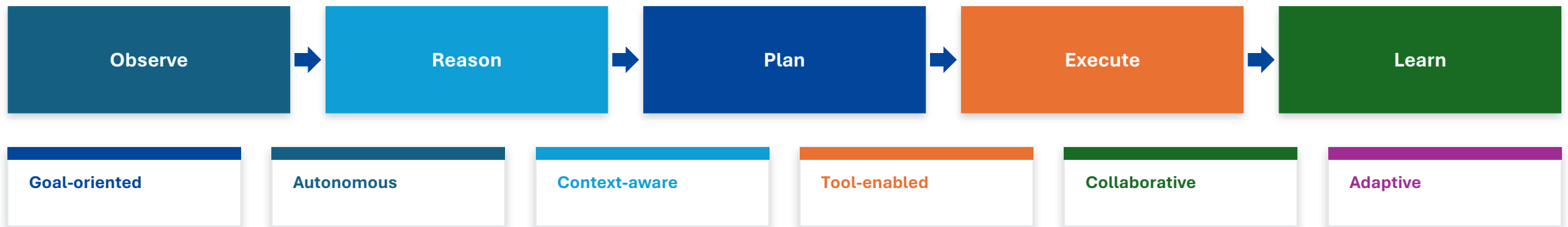
A refund once needed 5 employees across 3 departments.
An AI agent coordinates Service, Finance, Inventory, ERP and customer notifications — in minutes.

DID YOU KNOW?

Enterprise AI is shifting from chat-based assistants toward workflow automation — AI agents that perform multi-step tasks under human oversight.

What Is Agentic AI?

Agentic AI: intelligent software agents that reason, plan, act, use enterprise tools, and adapt toward defined business goals.



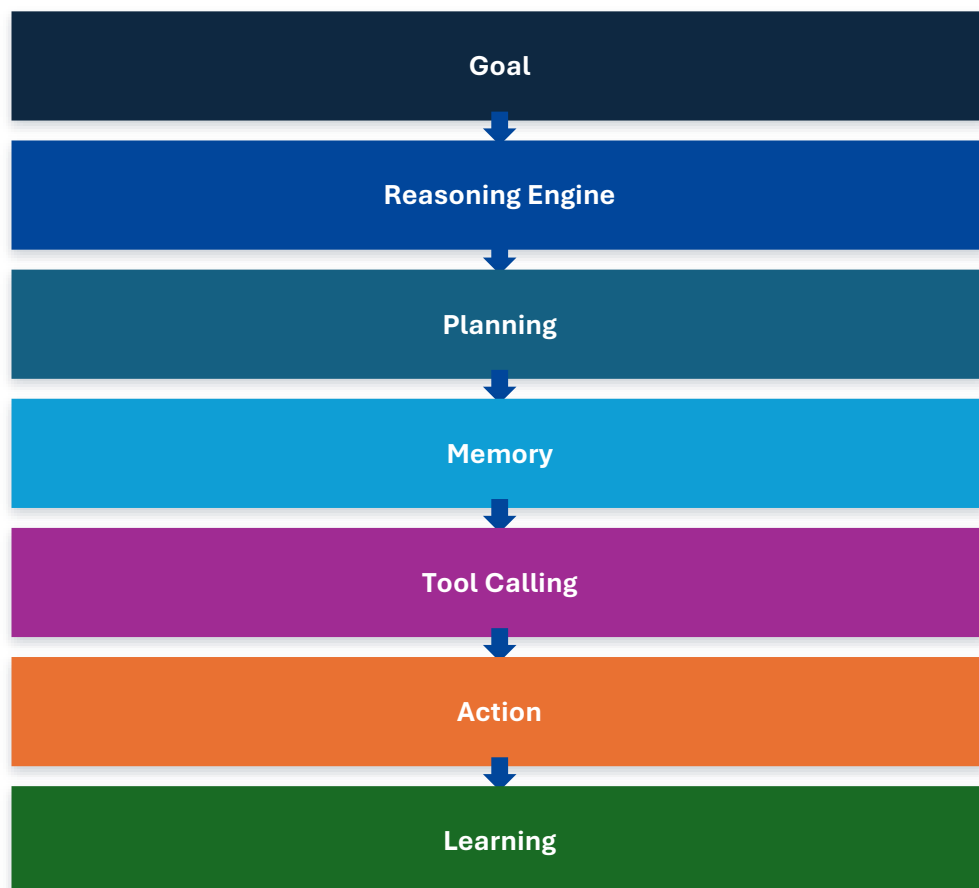
BUSINESS EXAMPLE — “I need a laptop before Monday.”

The agent checks inventory, creates an approval request, books delivery, notifies IT, and updates the employee.

Generative AI vs Copilot vs Agentic AI

Capability	Generative AI	Microsoft Copilot	Agentic AI
Answer questions	✓	✓	✓
Generate content	✓	✓	✓
Use enterprise data	Limited	✓	✓
Call business apps	—	Some	✓
Make decisions	—	Limited	✓
Complete workflows	—	Limited	✓
Work autonomously	—	—	✓

Anatomy of an AI Agent



Reasoning

Decides the next best step

Memory

Retains task context

Tools

Executes real actions

Knowledge

Retrieves trusted facts

Execution

Completes workflows

Human Feedback

Handles exceptions

INVOICE PROCESSING AGENT

Reads invoice → validates vendor → matches PO → requests approval → posts into ERP.

Single-Agent vs Multi-Agent Systems

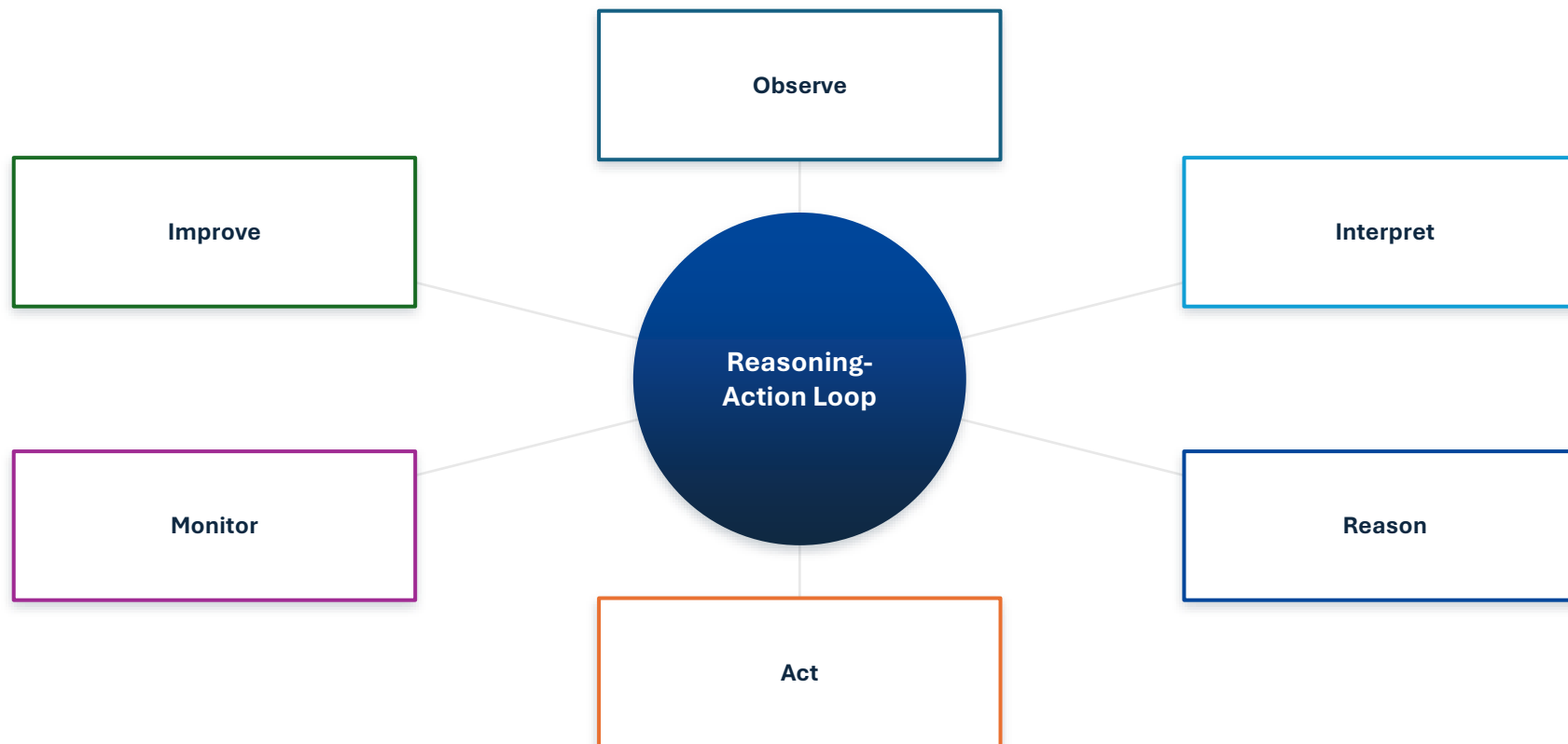


WHY MULTI-AGENT?

Specialized expertise • better scalability • parallel processing • independent reasoning • easier maintenance.

Refund example: Support → Finance → Fraud → Inventory → Notification agents collaborate end-to-end.

How Agentic AI Solves Business Problems



WHY IT MATTERS

Business conditions constantly change. Weather delays a shipment — the agent finds an alternate carrier, recalculates ETA, updates the warehouse, informs the customer, and revises the delivery plan. Agents adapt instead of deciding once.

Where Can Agentic AI Deliver Value?

Retail

Demand forecasting

Healthcare

Patient coordination

Banking

Loan processing

Manufacturing

Predictive maintenance

Supply Chain

Inventory optimization

HR

Employee onboarding

Finance

Invoice automation

IT

Service desk automation

COMMON BENEFITS

Faster execution • better decisions • lower operational costs • reduced manual effort • improved customer satisfaction.

Section 1 — Key Takeaways

01

Agentic AI goes beyond conversation to real business action.

02

AI agents reason, plan, and execute end-to-end workflows.

03

Microsoft provides an integrated ecosystem for enterprise AI.

04

Value comes from combining AI with enterprise data and applications.

WHAT'S COMING NEXT

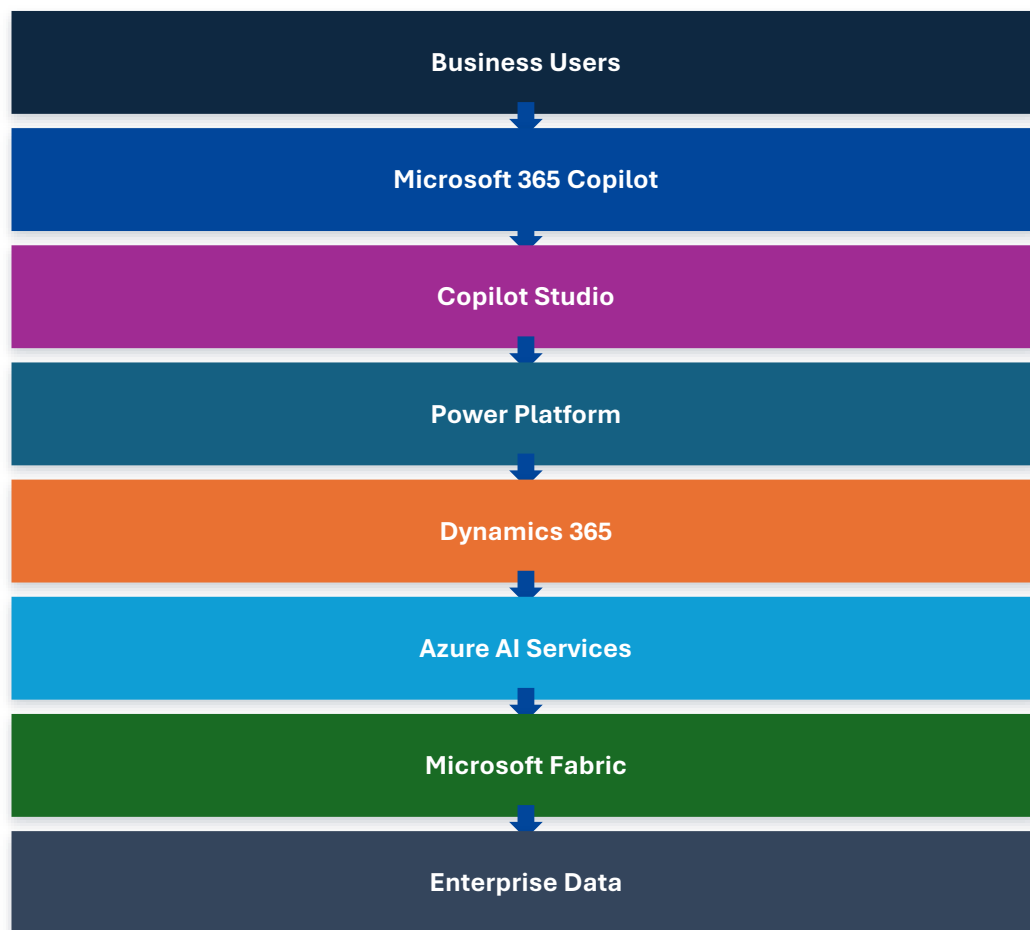
The Microsoft AI Ecosystem — Copilot, Copilot Studio, Power Platform, Dynamics 365, Azure AI & Fabric.

SECTION 02

The Microsoft AI Ecosystem



Microsoft AI Ecosystem at a Glance



Microsoft 365 Copilot

Productivity

Copilot Studio

AI agents

Power Platform

Automation

Dynamics 365

Business apps

Azure AI

Intelligence services

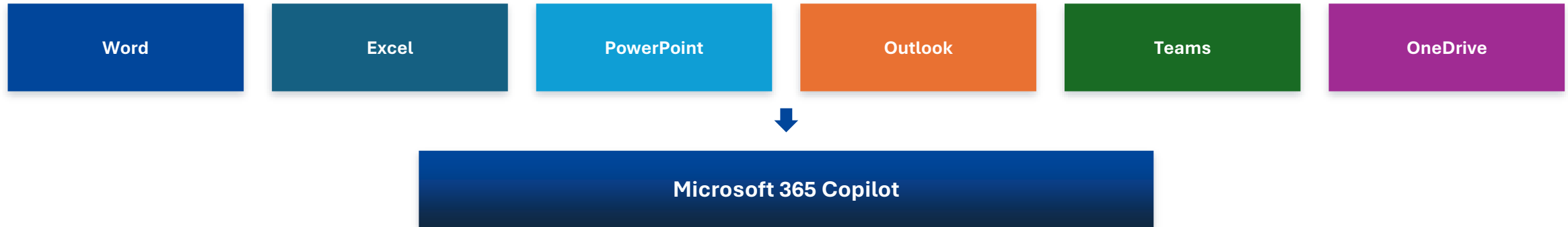
Microsoft Fabric

Unified data

ONE CONNECTED PLATFORM

Productivity, business apps, low-code, cloud AI, and data unified for enterprise AI.

Microsoft 365 Copilot — AI Productivity Assistant



Core Capabilities

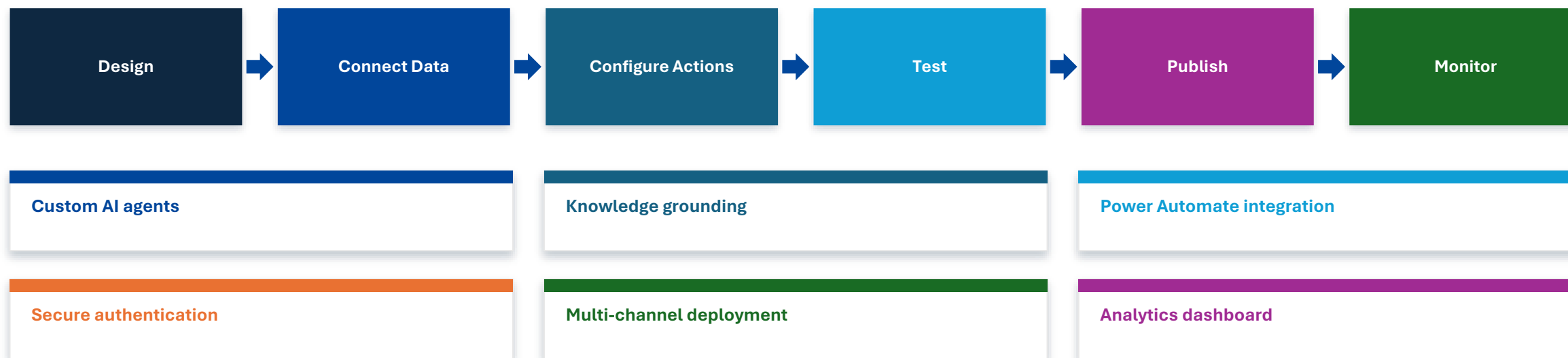
- Draft emails and documents
- Summarize Teams meetings
- Create presentations
- Analyze Excel data
- Search enterprise knowledge
- Prepare meeting briefs

BEST FOR

Knowledge work
Content generation
Meeting productivity
Document intelligence

Demo #2: build a presentation from a Word document.

Copilot Studio — Build Enterprise AI Agents



HR AI AGENT — “How many vacation days do I have?”

Checks the HR system, retrieves the leave balance, explains company policy, and initiates a leave request — all within organizational permissions.

Power Platform — Automating Business Processes

Power Apps

Build business applications

Power Automate

Workflow automation

Power BI

Business intelligence

Power Pages

External business portals

Unified by the Power Platform — low-code apps, AI-enhanced workflows, and Dataverse.

Submit receipt



AI extracts data



Policy validation



Manager approval



Finance updated

EXPENSE APPROVAL

Benefits: low-code development • AI-enhanced workflows • faster automation • business agility.

Dynamics 365 — AI Across Business Functions

Sales

Lead prioritization

Customer Service

Case summarization

Supply Chain

Inventory optimization

Finance

Cash flow forecasting

Field Service

Technician scheduling

Marketing

Personalized campaigns

BUSINESS SCENARIO — Customer contacts support

Dynamics 365 AI retrieves the customer profile, summarizes previous interactions, suggests the next best action, and generates a response draft.

Azure AI Services — Intelligence Behind Enterprise AI

Azure OpenAI

Large language models

AI Search

Enterprise retrieval

Document Intelligence

Extract structured data

Vision

Image understanding

Speech

Speech-to-text

Translator

Language translation

Documents • Images • Voice



Azure AI Services

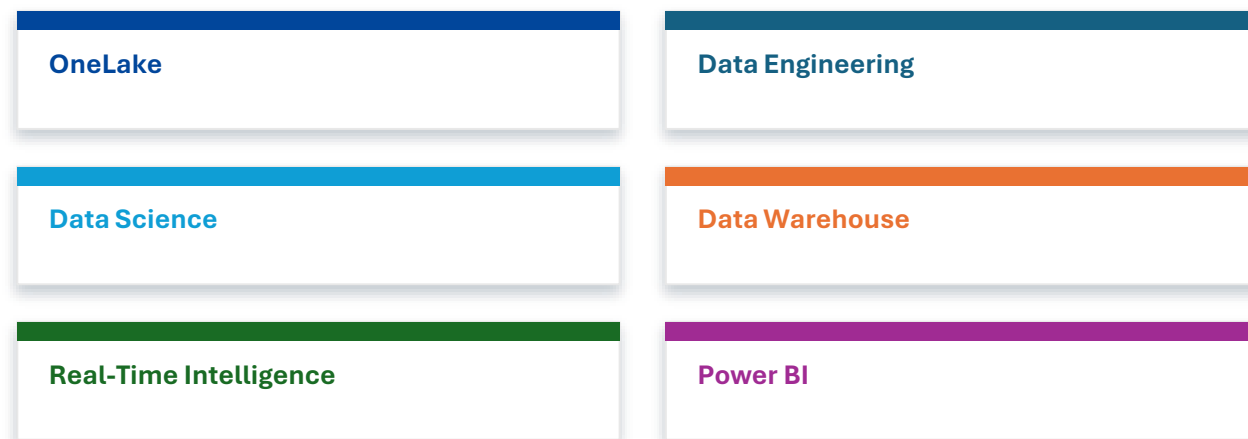
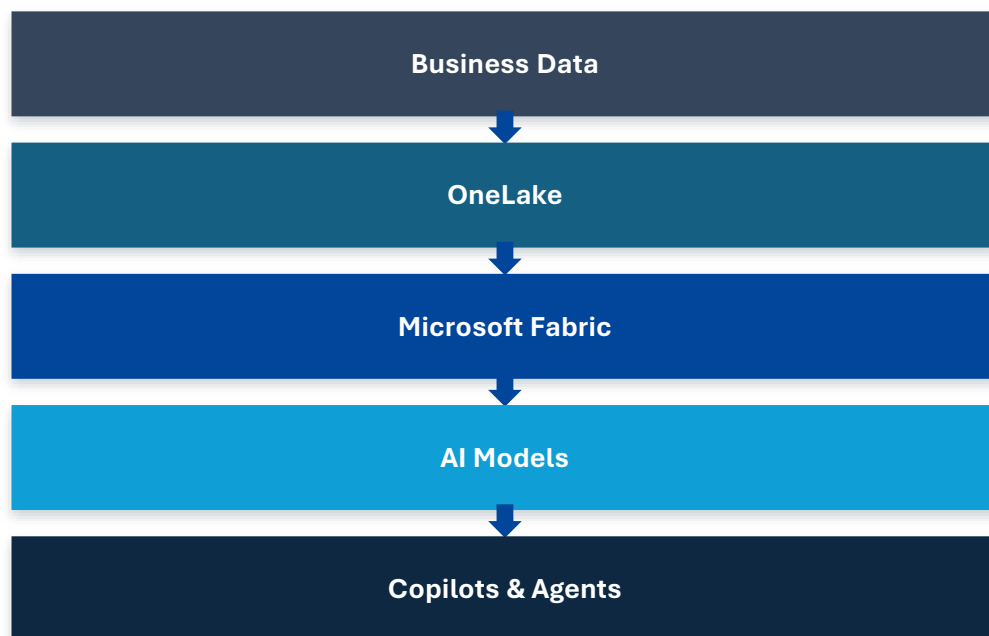


Business Applications

INVOICE PROCESSING

Upload invoice → Document Intelligence extracts fields → Azure OpenAI validates → ERP updated automatically. (Demo #3)

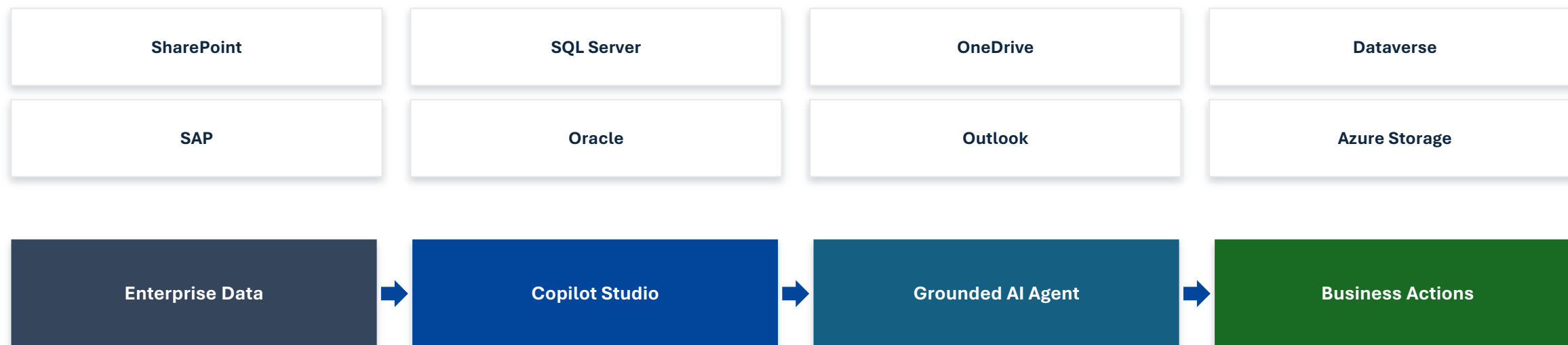
Microsoft Fabric — Data Foundation for AI



DID YOU KNOW?

Microsoft Fabric eliminates traditional data silos by bringing analytics, AI, and governance into a single SaaS platform — AI is only as good as the data it can access.

Connecting Enterprise Data Securely



KEY PRINCIPLE

AI should respond using authorized enterprise data — ensuring relevance, security, and compliance.

Legal agent: retrieves the latest SharePoint policy, related FAQs, and the expense form.

Choosing the Right Microsoft AI Tool

Business Need	Recommended Microsoft AI Tool
Draft documents	Microsoft 365 Copilot
Build AI agents	Copilot Studio
Automate workflows	Power Automate
CRM intelligence	Dynamics 365
Enterprise AI models	Azure AI
Unified analytics	Microsoft Fabric

CHOOSE BY OUTCOME

Select the tool based on the business outcome you need — not the technology alone.

Section 2 — Key Takeaways

01

Microsoft offers a complete AI platform — not isolated tools.

02

Each service addresses a different business need.

03

AI succeeds when productivity, automation, apps and data work together.

04

The right tool depends on the business outcome, not the technology.

WHAT'S COMING NEXT

Architecting Enterprise Solutions — reference architecture, design patterns and grounding.

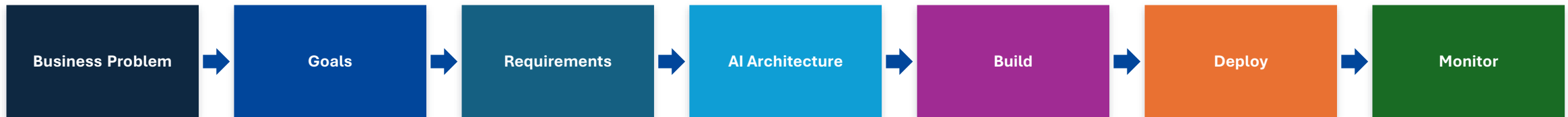
SECTION 03

Architecting Enterprise Solutions



From Business Challenge to AI Solution

Start with **“What business outcome are we trying to achieve?”** — not “Which AI tool should we use?”



Solution Design Questions

- What process needs improvement?
- What decisions should AI support?
- Which tasks can be automated?
- Where is human approval required?
- How will success be measured?

BUSINESS EXAMPLE

Challenge: support response time is 12 hours.

AI goal: reduce it to under 2 hours.

Solution: triage tickets, search knowledge, draft responses, escalate complex cases.

Identifying Business & Technical Requirements

Business Requirements

- Business objectives
- KPIs & success metrics
- User experience
- Compliance needs
- Cost targets

Technical Requirements

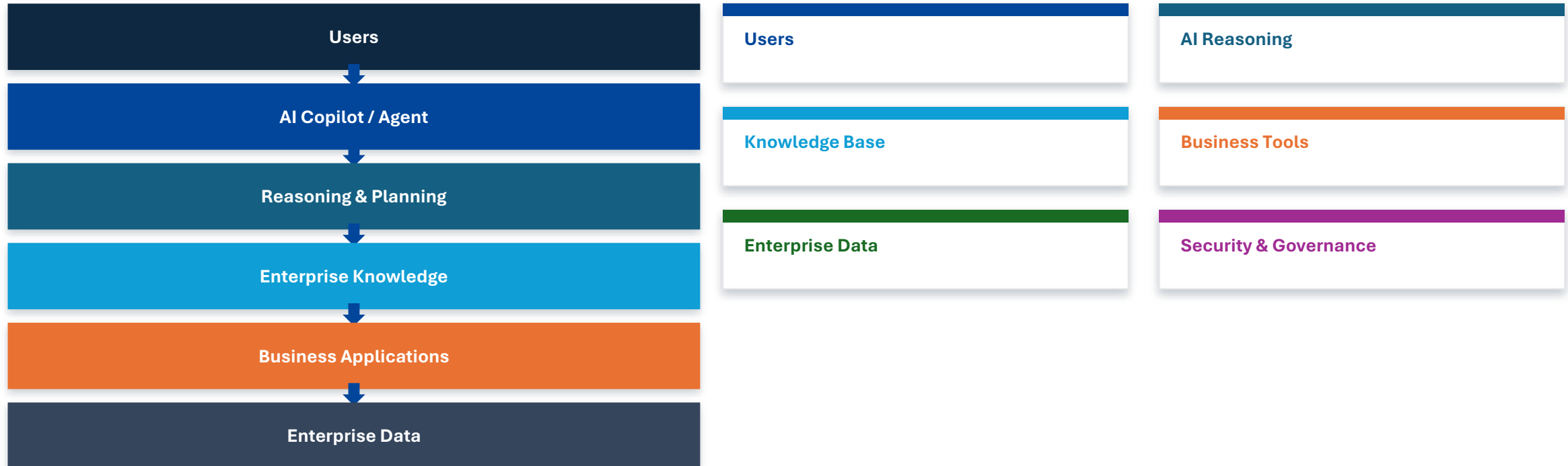
- Data sources
- Integrations
- Identity & access
- Scalability
- Performance
- Availability

Business Goals + Technical Capabilities = Successful AI Solution

EXAMPLE

Employees spend hours searching policies → SharePoint knowledge, a Copilot Studio agent, Azure AI Search, and Microsoft Entra ID authentication.

Designing an Enterprise AI Architecture

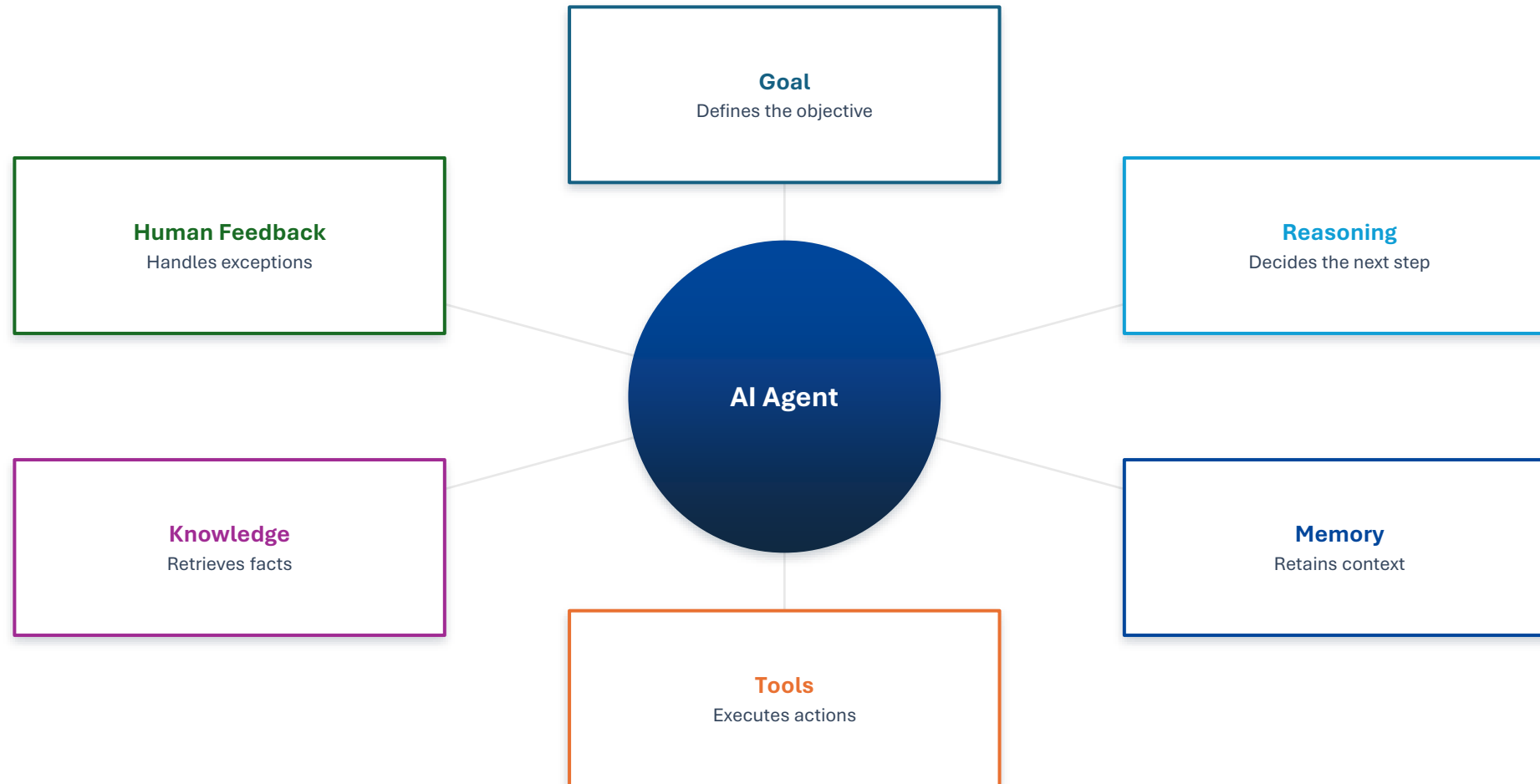


DESIGN PRINCIPLE

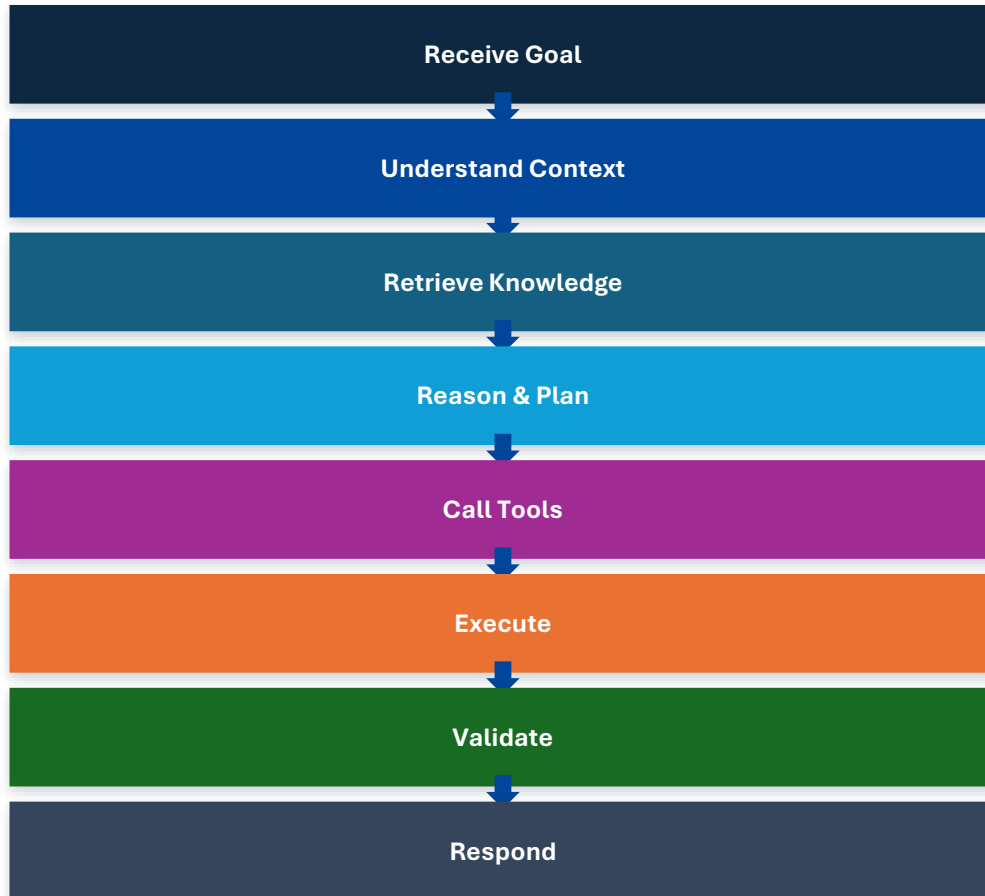
Separate reasoning, knowledge, and actions to build modular, scalable AI solutions.

Procurement agent: understands requests, retrieves contracts, checks approval rules, creates the PO, notifies stakeholders.

Anatomy of an AI Agent — Six Core Capabilities



Designing Intelligent Agent Workflows



Workflow Best Practices

- Break large tasks into smaller steps
- Validate outputs before execution
- Log every action
- Handle failures gracefully
- Escalate uncertainty

“Book travel for my conference.”

Reads the travel policy
Finds flights
Creates the itinerary
Requests manager approval
Sends a calendar invite
(Demo #5: agent invokes Power Automate)

Grounding AI with Enterprise Knowledge



“What is our parental leave policy?”

The agent searches SharePoint, retrieves the latest HR policy, summarizes only approved content, and links to the source document.

DID YOU KNOW?

Grounding with enterprise data reduces hallucinations and keeps AI responses aligned with organizational knowledge and permissions.

Memory & Context Management

Conversation Memory

Current interaction

Task Memory

Workflow progress

Enterprise Memory

Policies & documents

User Context

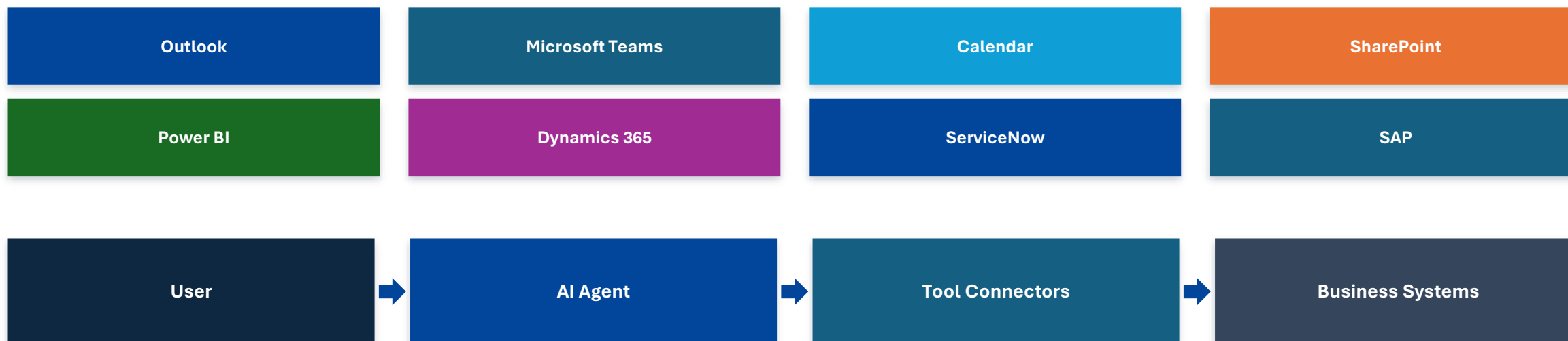
Role, permissions, preferences

Without context, every interaction starts from scratch. With memory, agents continue work, personalize responses, and stop asking for the same information.

BUSINESS EXAMPLE

An employee pauses a travel request. The next day the agent resumes — destination, preferred airline, budget, and pending approvals — without asking again.

Connecting AI to Enterprise Tools

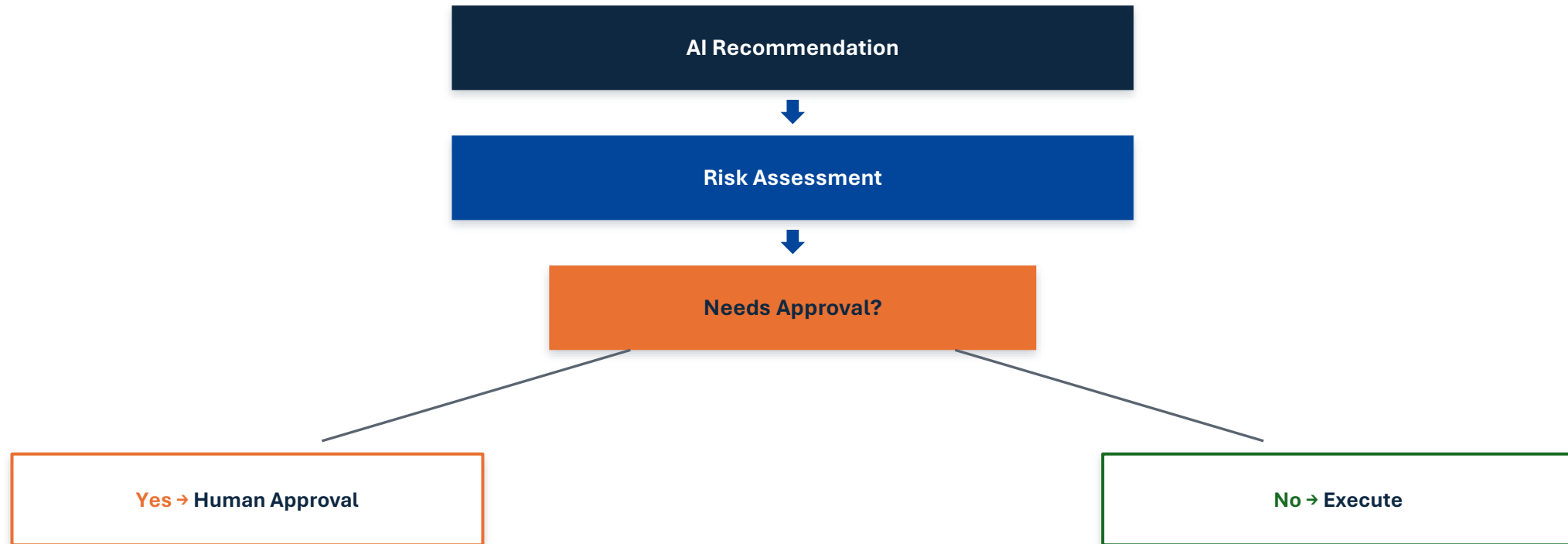


TYPICAL ACTIONS — Sales Agent

Send email • create ticket • update CRM • schedule meeting • generate report • start workflow.

Creates the opportunity in Dynamics 365, schedules a follow-up, emails the proposal, and updates the CRM automatically.

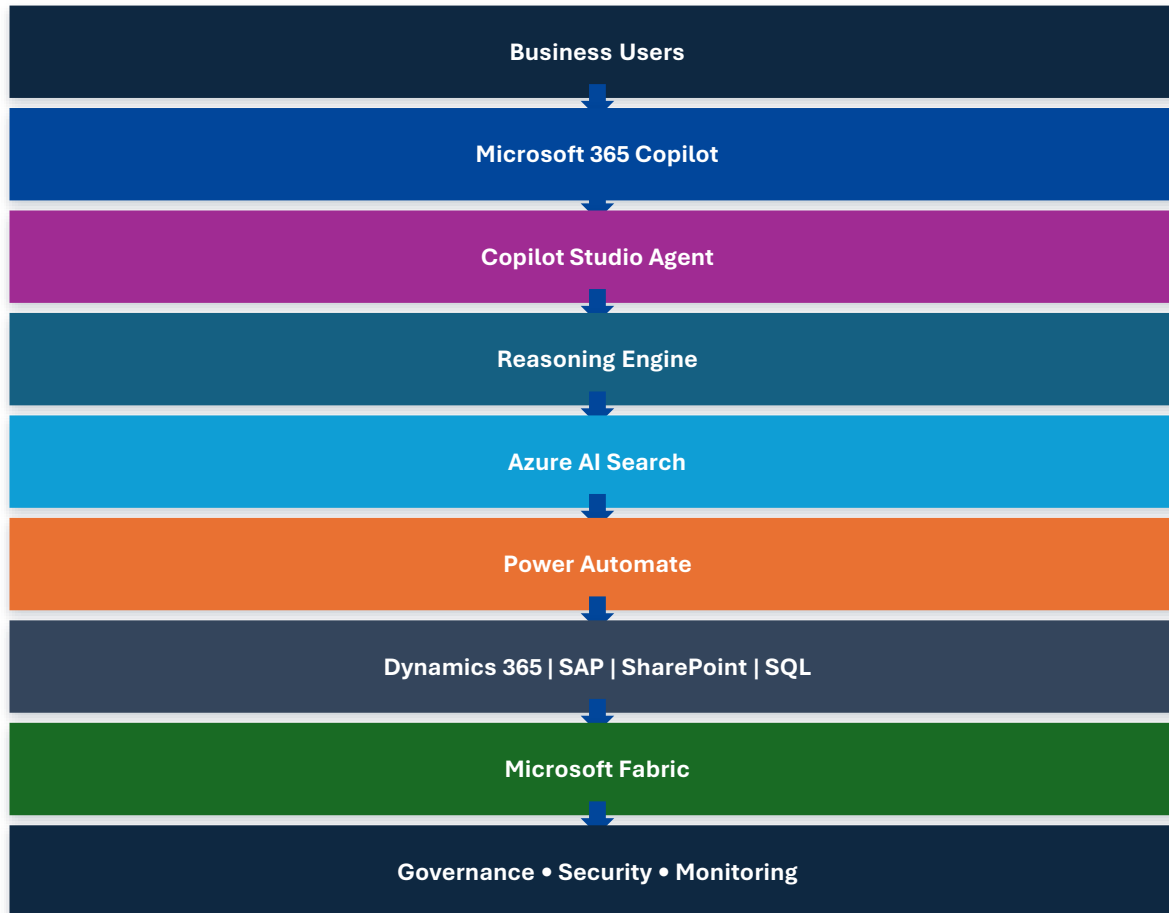
Human-in-the-Loop Design



WHEN HUMAN APPROVAL IS ESSENTIAL

High-value financial transactions • legal decisions • clinical recommendations • HR policy exceptions • security-sensitive actions.

Enterprise Architecture Blueprint



Architecture Principles

- Business-first design
- Modular AI services
- Grounded enterprise knowledge
- Secure integrations
- Human oversight
- Continuous monitoring

CAPSTONE — Onboarding Agent

Goal: cut onboarding from five days to one.

Retrieves the checklist, creates IT accounts, orders a laptop, schedules orientation, assigns training, and updates the HR dashboard.

Outcome: 80% fewer manual tasks, faster onboarding, consistent experience, improved compliance.

SECTION 04

Governance, Security & Scaling



Building Responsible Agentic AI

Fairness

Reduce bias, ensure equitable outcomes

Privacy

Protect sensitive business & customer data

Security

Prevent unauthorized access and misuse

Transparency

Explain how AI reaches decisions

Accountability

Maintain human ownership of outcomes

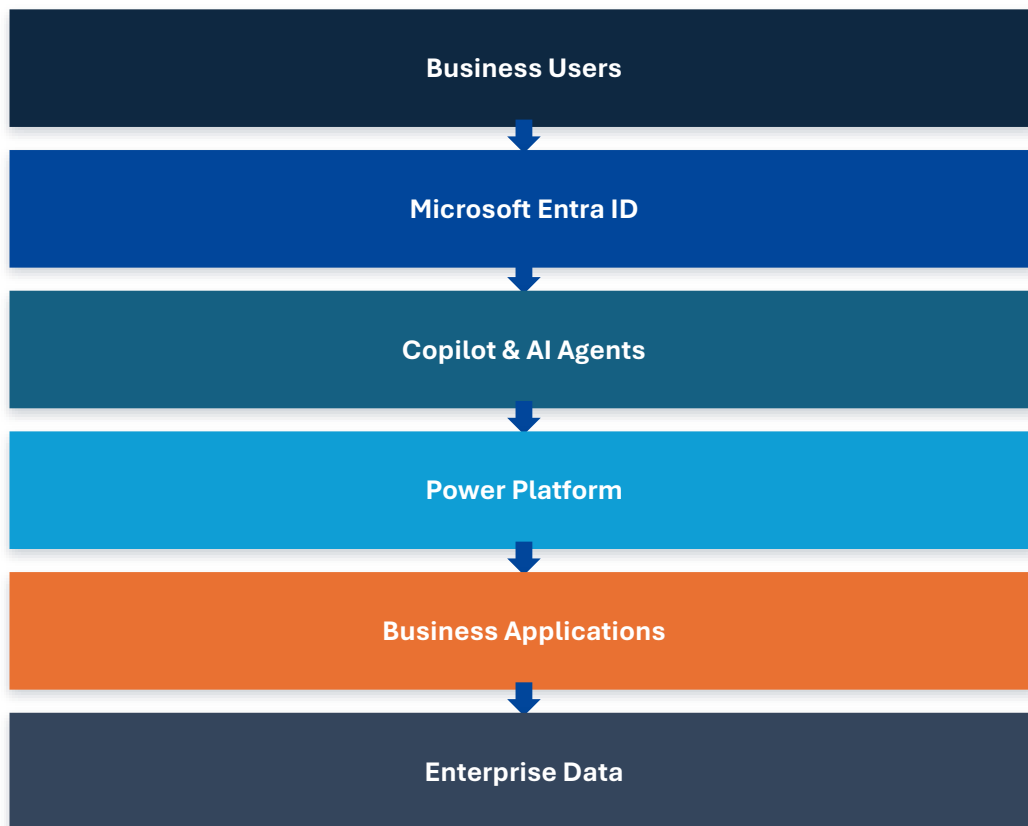
Reliability

Deliver consistent, dependable performance

BUSINESS EXAMPLE — Loan Recommendation Agent

Provides a decision summary, supporting business rules, source-data references, and confidence indicators — so loan officers review and approve responsibly.

Securing Enterprise AI



Microsoft Entra ID
Identity & authentication

Microsoft Purview
Data governance & compliance

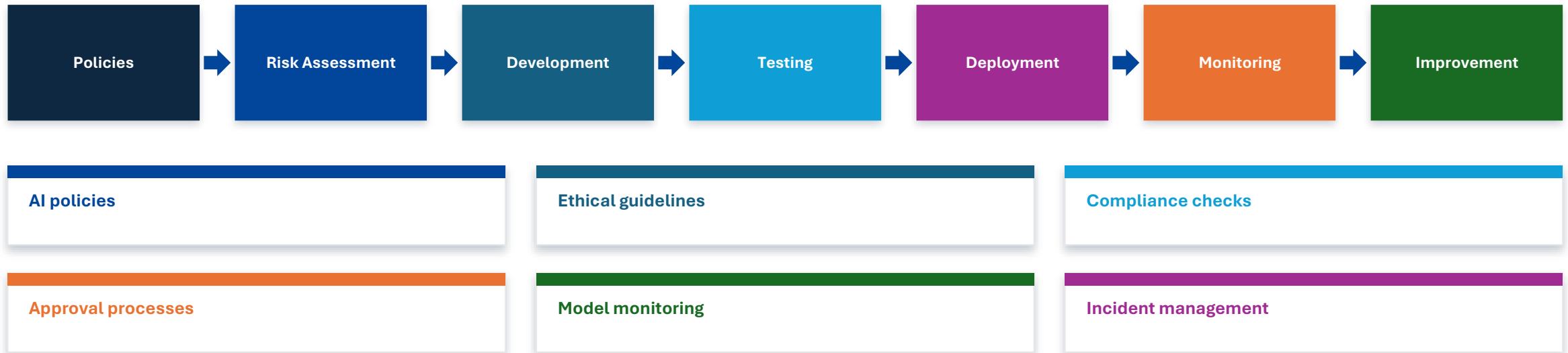
Microsoft Defender
Threat protection

Microsoft Sentinel
Security monitoring & response

BEST PRACTICES

MFA • least-privilege access • encryption • audit logging • continuous monitoring.

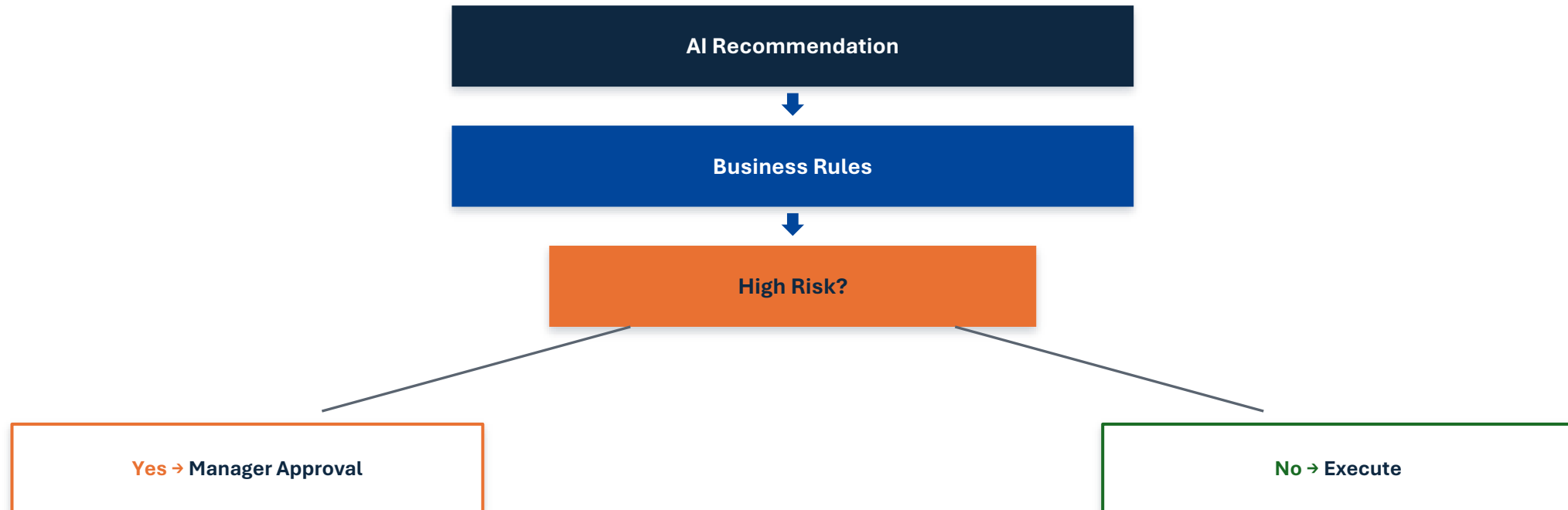
AI Governance Framework



BEFORE DEPLOYING A PROCUREMENT AGENT

Security review completed, compliance approved, human approval thresholds configured, and monitoring dashboards enabled.

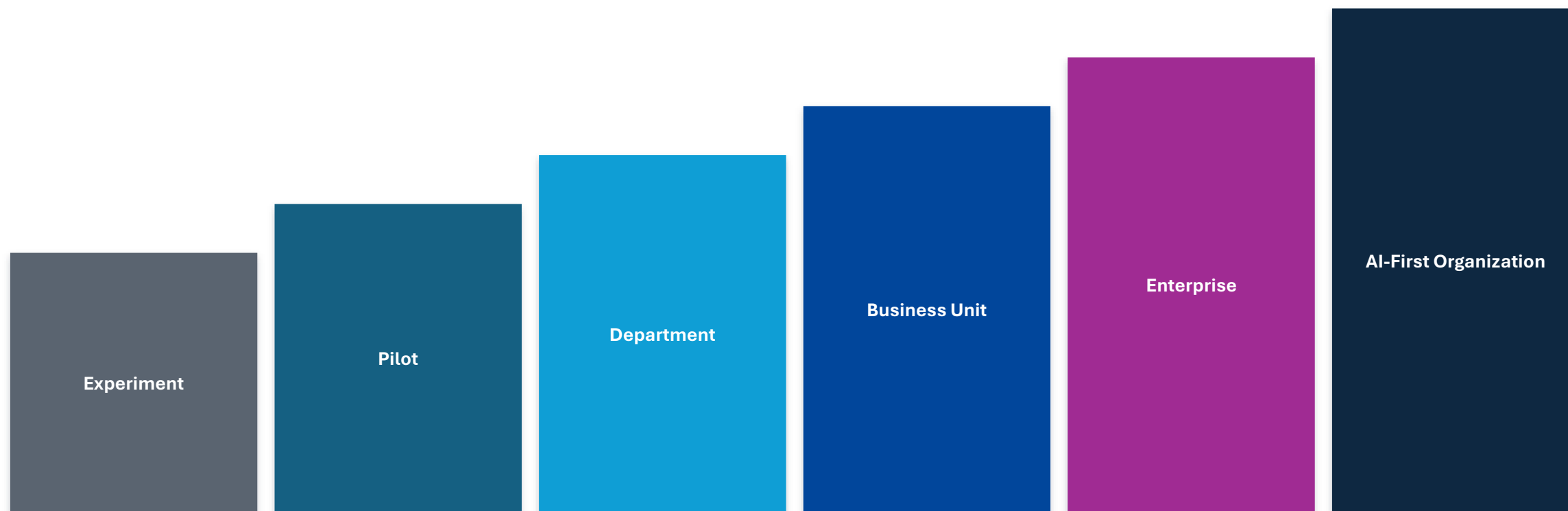
Human-in-the-Loop Governance



TYPICAL APPROVAL SCENARIOS

Payments above threshold • contract approvals • HR policy exceptions • clinical recommendations • legal decisions.
Automate repetitive work, but retain human judgment for sensitive, ethical, or high-value decisions.

Scaling Agentic AI Across the Enterprise



SUCCESS FACTORS

Executive sponsorship • business ownership • reusable AI components • governance • user training • change management. Start with HR onboarding, expand to IT, finance, procurement, and service.

Common Challenges & Best Practices

Challenge	Recommended Practice
Hallucinations	Ground with enterprise knowledge
Poor data quality	Clean and govern data
Security risks	Apply Zero Trust & RBAC
Shadow AI	Define AI usage policies
Low adoption	Train users and communicate value
Workflow failures	Monitor and improve continuously
DID YOU KNOW? Most production AI failures come from process, governance, or data quality — not the AI model itself. Ground responses in SharePoint, cite sources, and escalate uncertain answers to a human.	

Future of Agentic AI with Microsoft



Emerging Trends

- Multi-agent collaboration
- Smarter reasoning models
- Cross-platform orchestration
- AI-driven decision intelligence
- Continuous learning systems

BUSINESS VISION

“Improve next month's customer satisfaction by 10%.”

A network of agents analyzes feedback, updates workflows, coordinates departments, tracks progress, and reports outcomes.

THE NEXT WAVE

Enterprise AI is moving from single copilots to coordinated, multi-agent systems that continuously learn and act across the business.

Executive Takeaways — 10 Key Lessons

Start with outcomes

Not the technology

Design modular

Reusable AI architectures

Ground with knowledge

Trusted enterprise data

Connect to apps

Business applications

Keep humans involved

Critical decisions

Build security in

Every layer

Govern the lifecycle

End to end

Measure ROI

Business value

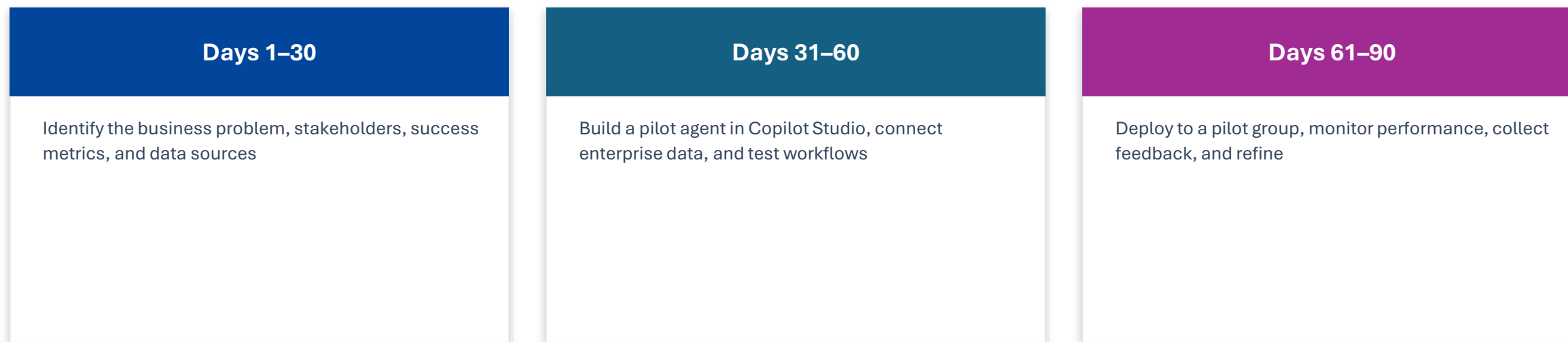
Scale gradually

Proven use cases

People + technology

Succeed together

Your Agentic AI Implementation Roadmap



RECOMMENDED FIRST PROJECTS

Employee onboarding • IT help desk • policy assistant • sales proposal generation • invoice processing.

Demo





The Next Step:

**Continue your Learning &
Get Microsoft Certified**



Getting Microsoft Certified:

What are the benefits of certification?

Stand out

Certified employees **get greater recognition** of skills due to validation¹

Earn more

23% of Microsoft certified technologists **earn up to 20% more**²

Advance

49% believe cloud certifications **increase employability**³

¹ 2017 IDC-Microsoft, [Cloud Skills and Organizational Influence: How Cloud Skills Are Accelerating the Careers of IT Professionals](#) white paper

² 2017 Pearson VUE, [Value of Certification](#) white paper

³ 2017 [Nigel Frank International Microsoft Technology for Business Salary Survey](#)

Broadcast certified skills

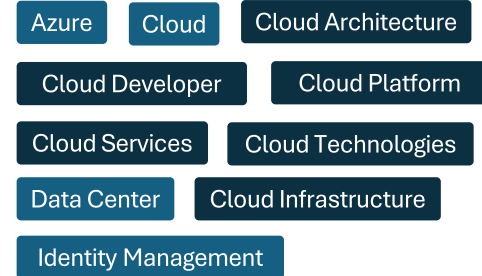


Instant employer verification



This badge is issued by Microsoft

Skills



Unlock the value of **Microsoft Open Badges**

- **Social** - help you share your skills on the social platform of your choosing and your own personal communication
- **Discoverable** – helps people find you and employers who may be interested
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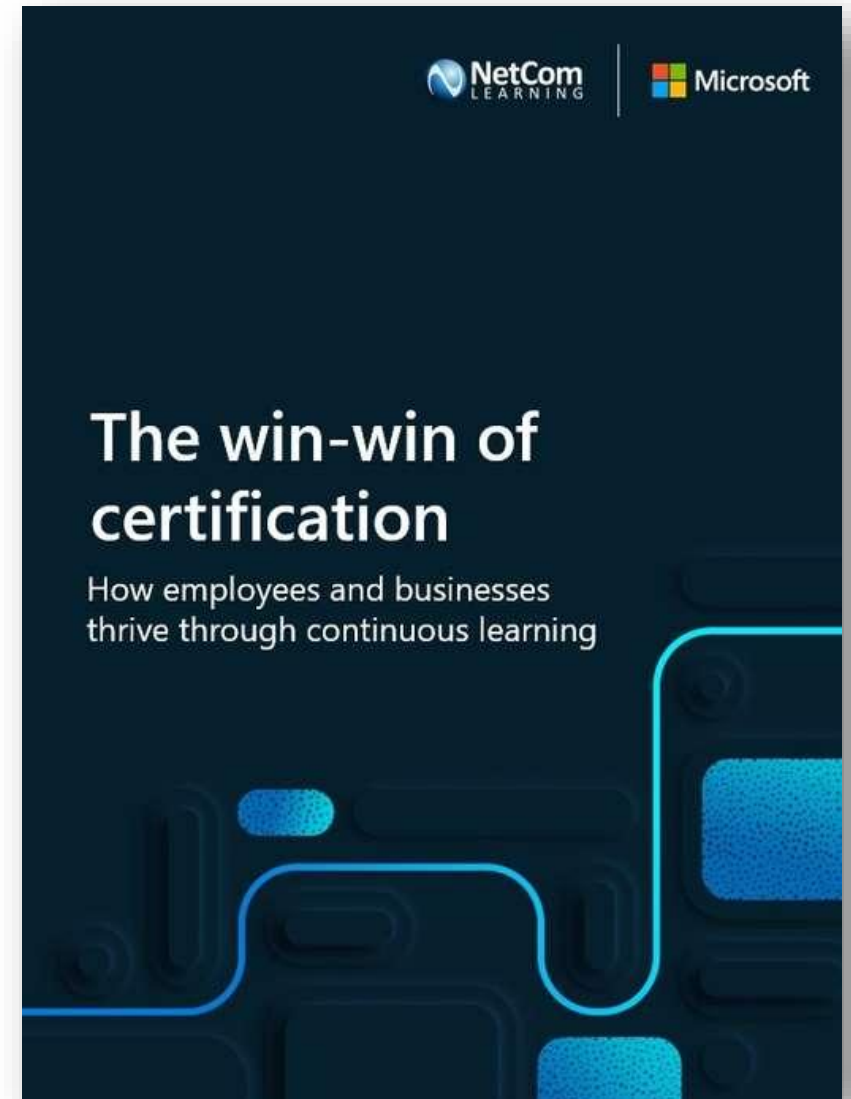
Explore the eBook:

Let the data speak

When organizations maintain a culture of certification and continuous learning, the results are both substantial and measurable: **well-trained and certified teams are almost 20% more productive than other staff.**

Quantify the value of a growth mindset and training. In this eBook, you'll see stats and facts on how certification benefits both employees and organizations, including:

- How training improves timeliness, insight, and business impact
- The personal and career benefits of certification
- The direct relationship between continuous learning and higher pay
- How certification improves productivity, efficiency, and job satisfaction



Download Link is in your Event Handout

Grow your business & career with Official Microsoft **Role-Based** Instructor-led Classroom Training



Key Highlights:

- Highly interactive discussions on your specific **real-world use cases** and business contexts with **Microsoft Certified Trainers** who come with technical subject matter expertise
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Some facts about Microsoft Certification Renewals

Benefits of renewing your certification

- Maintain your Microsoft Certification with a continuous learning mindset and keep your skills current at no cost.
- Validate your skills with the latest technology updates that affect the job role.
- Renew your certification without having to pay for and pass any exam(s).

Certification renewal requirements

- If you hold a valid Microsoft role-based or specialty certification that expires within approximately six months, you may renew it.
- You will receive an email from Microsoft Certification team 6 months prior to your certification expiration date, once your certification is eligible for renewal.
- To renew your certification, pass the associated assessment on Microsoft Learn *before* your certification expires.
- Passing the renewal assessment on Microsoft Learn is the only way to renew a certification. You cannot renew your certification by passing an exam, even if it's a new or updated version of the exam you initially passed to earn the certification.

Get in touch with NetCom Learning for end-to-end guidance and support on your Microsoft training and certification requirements



Let's Get Trained & Certified

 **Microsoft**
Solutions Partner
Microsoft Cloud

Training Services

Questions

Redeem Your Courseware & Certificate

Thank You for Attending!

Architecting Agentic AI Business Solutions with Microsoft AI

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